

CS4084

Quantum Computing

Week 03 – Lecture 05

Dr. Maqsood M. Khan

Department of Computer Science
FAST-NUCES Peshawar

Outline

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Quick Review: What We Covered Last Week

- **Quantum Foundations:** Introduction to Quantum Information, Computation, and its physical description.
- **Classical vs. Quantum:** Review of classical information processing and its mathematical representation.
- **Mathematical Framework:**
 - Introduction to **Dirac Notation** (Bra-ket formalism).
 - Worked mathematical examples of state vectors.
- **System Dynamics:**
 - Measuring probabilistic states.
 - **Deterministic Operations:** Classical operations on a system and their mathematical examples.

Key Message: Understanding the transition from deterministic classical operations to probabilistic quantum states is essential for mastering quantum algorithms.

Dirac Notation (second part)

Let Σ be any classical state set, and assume the elements of Σ have been placed in correspondence with the integers $1, \dots, |\Sigma|$.

We denote by $\langle \alpha |$ the **row vector** having a 1 in the entry corresponding to $\alpha \in \Sigma$, with 0 for all other entries.

Example

If $\Sigma = \{0, 1\}$, then

$$\langle 0 | = \begin{pmatrix} 1 & 0 \end{pmatrix} \quad \text{and} \quad \langle 1 | = \begin{pmatrix} 0 & 1 \end{pmatrix}$$

Dirac Notation (second part)

Multiplying a row vector to a column vector yields a scalar:

$$\begin{pmatrix} * & * & * & \dots & * \end{pmatrix} \begin{pmatrix} * \\ * \\ * \\ \vdots \\ * \end{pmatrix} = (*)$$

$$\begin{pmatrix} 0 & 1 & 0 & \dots & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = (0)$$

$$\langle a|b \rangle = \langle a||b \rangle = \begin{cases} 1 & a = b \\ 0 & a \neq b \end{cases}$$

Example Check for other possibilities for

Let Σ be any classical state set, and assume the elements of Σ have been placed in correspondence with the integers $1, \dots, |\Sigma|$.

We denote by $\langle \alpha |$ the **row vector** having a 1 in the entry corresponding to $\alpha \in \Sigma$, with 0 for all other entries.

Example

If $\Sigma = \{0, 1\}$, then

$$\langle 0 | = \begin{pmatrix} 1 & 0 \end{pmatrix} \quad \text{and} \quad \langle 1 | = \begin{pmatrix} 0 & 1 \end{pmatrix}$$

Dirac Notation (second part)

Multiplying a column vector to a row vector yields a matrix:

$$\begin{pmatrix} * \\ * \\ * \\ \vdots \\ * \end{pmatrix} \begin{pmatrix} * & * & * & \dots & * \end{pmatrix} = \begin{pmatrix} * & * & * & \dots & * \\ * & * & * & \dots & * \\ * & * & * & \dots & * \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ * & * & * & \dots & * \end{pmatrix}$$

Example

$$|0\rangle\langle 0| = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

Outer Products in Dirac Notation

- $|0\rangle \langle 1|$:

$$|0\rangle \langle 1| = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

- $|1\rangle \langle 0|$:

$$|1\rangle \langle 0| = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

- $|1\rangle \langle 1|$:

$$|1\rangle \langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

Deterministic Operations

Every function $f : \Sigma \rightarrow \Sigma$ describes a **deterministic operation** that transforms the classical state α into $f(\alpha)$, for each $\alpha \in \Sigma$.

Given any function $f : \Sigma \rightarrow \Sigma$, there is a (unique) matrix M satisfying

$$M |\alpha\rangle = |f(\alpha)\rangle \quad (\text{for every } \alpha \in \Sigma)$$

This matrix may be expressed as

$$M = \sum_{b \in \Sigma} |f(b)\rangle \langle b|$$

Its action on standard basis vectors works as required:

$$M |\alpha\rangle = \left(\sum_{b \in \Sigma} |f(b)\rangle \langle b| \right) |\alpha\rangle = \sum_{b \in \Sigma} |f(b)\rangle \langle b | \alpha \rangle = |f(\alpha)\rangle$$

Probabilistic operations

Probabilistic operations are classical operations that may introduce randomness or uncertainty.

Example

Here is a probabilistic operation on a bit:

If the classical state is 0, then do nothing.

If the classical state is 1, then flip the bit with probability $1/2$.

Probabilistic operations are described by **stochastic matrices**:

- All entries are nonnegative real numbers
- The entries in every column sum to 1

Example (Probabilistic Operations on a bit):

$$\begin{pmatrix} 1 & \frac{1}{2} \\ 0 & \frac{1}{2} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Composing Operations

Suppose X is a system and $M_1, \dots, M_n$ are stochastic matrices representing probabilistic operations on X .

Applying the first probabilistic operation to the probability vector v , then applying the second probabilistic operation to the result yields this vector:

$$M_2(M_1 v) = (M_2 M_1) v$$

The probabilistic operation obtained by *composing* the first and second probabilistic operations is represented by the *matrix product* $M_2 M_1$.

Composing the probabilistic operations represented by the matrices $M_1, \dots, M_n$ (in that order) is represented by this matrix product:

$$M_n \cdots M_1$$

Ordering does matters!

The order is important: matrix multiplication is *not commutative!*

$$M_1 = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$$

$$M_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$M_2 M_1 = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$$

$$M_1 M_2 = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$$

Quantum Information

A **quantum state** of a system is represented by a **column vector** whose indices are placed in correspondence with the classical states of that system:

- The entries are complex numbers.
- The sum of the absolute values squared of the entries must equal 1.

Definition

The **Euclidean norm** for vectors with complex number entries is defined like this:

$$v = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} \implies \|v\| = \sqrt{\sum_{k=1}^n |\alpha_k|^2}$$

Quantum state vectors are therefore **unit vectors** with respect to this norm.

Examples of qubit states

- Standard basis states: $|0\rangle$ and $|1\rangle$
- Plus/minus states:

$$|+\rangle = \frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \quad \text{and} \quad |-\rangle = \frac{1}{\sqrt{2}} |0\rangle - \frac{1}{\sqrt{2}} |1\rangle$$

- A state without a special name:

$$\frac{1+2i}{3} |0\rangle - \frac{2}{3} |1\rangle$$

Summary: Key Takeaways

- **Mathematical Language:** Mastered **Dirac Notation** ($|\psi\rangle$) to represent system states compactly and calculate inner products.
- **Transition Logic:**
 - **Deterministic Operations:** Fixed transitions where one state leads to exactly one other state.
 - **Probabilistic Operations:** Systems where transitions are governed by stochastic matrices.
- **Composing Operations:** Learned how to combine multiple steps using matrix multiplication to determine final system outcomes.
- **Quantum Shift:** Unlike classical uncertainty (lack of knowledge), **Quantum Information** uses:
 - **Superposition:** States existing simultaneously.
 - **Complex Amplitudes:** The fundamental building blocks of quantum probability.

Next: We will apply these linear algebra foundations to **Unitary Transformations** and the physical realization of qubits.